

Statistical Inference on Aspirin

Leo Rogers

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1 Introduction

We will explore the International Stroke Trial (IST), a large randomised clinical trial assessing the effects of heparin and aspirin outcomes following a suspected acute ischaemic stroke. The trial consists of 19,435 patients from 467 hospitals in 36 countries. Our analysis considers 19,279 patients who have complete outcome data. Patients are enrolled into the trial within 48 hours of symptom onset and randomised into one of six treatment groups formed in a factorial design by two aspirin treatments Yes/No and three heparin treatments Medium/Low/None. This report focuses on evaluating aspirin affects on the probability of death at six months. The dataset contains patient characteristics recorded at baseline, treatment allocation indicators, and six month outcomes. Randomisation is expected to balance baseline characteristics across treatment groups so comparing these variables by aspirin treatment provides a useful initial assessment of whether groups appear comparable.

The IST employed a factorial design with independent randomisation to aspirin and heparin treatment; we include all patients regardless of heparin status to preserve randomisation and maximize statistical power.

Summary statistics are presented in Table 1. Both the No Aspirin and Aspirin groups consist of an approximately equal number of participants (≈ 9717). The groups were remarkably well balanced in terms of age with an equal mean of 71.7 years and standard deviation of 11.6 in each group. Systolic blood pressure means were also similar at 160.0 mmHg and standard deviation approximately 27.6. The proportion of male participants was 54.1% in the No Aspirin group vs. 53.0% in the Aspirin group. The conscious state of patients at randomisation are also very similar, the proportions being approximately 77% for fully alert, 22% drowsy and 1% unconscious in both groups.

Treatment Group	n	AGE: Mean (SD)	SBP: Mean (SD)	MALE: n (%)	F (%)	D (%)	U (%)
No Aspirin	9715	71.7 (11.6)	160.0 (27.5)	5255 (54.1)	76.8	21.9	1.34
Aspirin	9720	71.7 (11.6)	160.0 (27.7)	5152 (53.0)	76.8	21.9	1.34

Table 1: Summary statistics by treatment group. Age in years, systolic blood pressure in mmHg. Conscious state at randomisation: F = fully alert, D = drowsy, U = unconscious.

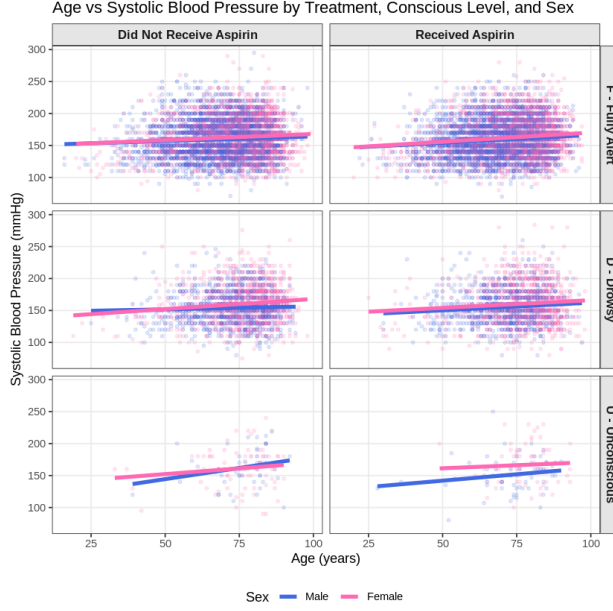


Figure 1: Joint distribution of age and baseline systolic blood pressure by treatment group and conscious state

Figure 1 shows the joint distribution of age and baseline systolic blood pressure at the time patients are randomised, split between male and female and grouped by treatment and conscious state at randomisation. Between each group the linear relationships appear similar across treatment and conscious groups with no obvious difference in location or spread between sexes.

Examination of Table 1 and Figure 1 shows that the treatment groups are well balanced at baseline, with comparable distributions of age, systolic blood pressure, sex, and conscious state.

2 Effects of Age, Sex, and Aspirin Treatment on Six-Month Mortality

We investigate a generalized linear model to fit to our data. Figure 2 shows the relationship between the mean proportion of deaths and the variance of these proportions averaged over months, grouped by age and country. The scatter plot exhibits a noisy quadratic pattern, with variance increasing as the mean proportion approaches 0.5 and decreasing as the mean approaches 0 or 1. This pattern is consistent with the mean and variance relationship of a binomial distribution, where the variance of a proportion is given by $p(1 - p)/n$. The observed shape therefore suggests that a binomial generalized linear model would be appropriate for modelling the probability of death, as it naturally accounts for the dependence of the variance on the mean. Minor deviations from the theoretical curve likely reflect sampling variability due to differing monthly sample sizes and random noise.

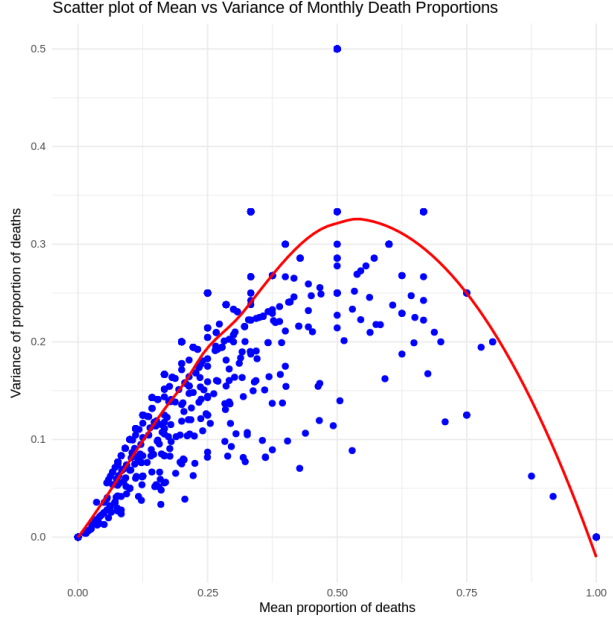


Figure 2: Mean proportion of deaths against variance of proportion of deaths

Table 2: Odds ratios from the logistic regression model predicting death at six months. Reference levels have odds ratio = 1.

Covariate	Odds Ratio
Intercept	0.010
Age (per year)	1.069
Sex (Female, reference)	1
Sex (Male vs Female)	1.090
Systolic BP (per mmHg)	0.997
Conscious (Drowsy, reference)	1
Conscious (Fully Alert vs Drowsy)	0.224
Conscious (Unconscious vs Drowsy)	4.340
Aspirin (No, reference)	1
Aspirin (Yes vs No)	0.940

After fitting a binomial generalized linear model, Table 2 presents the estimated odds ratios from the model predicting death at six months, with age, sex, systolic blood pressure, conscious state at randomization, and aspirin treatment being our covariates. We use a logistic link function so computing our odds ratios is done by exponentiating the coefficients from our regression model. The odds ratios for categorical variables are presented relative to their reference levels Female for sex, Drowsy for conscious, and No for Aspirin treatment. The results indicate that there is an increase in odds of death with each year increase in age which is expected, whereas higher systolic blood pressure is associated with slightly lower odds of death. Conscious state has a substantial effect where patients in an unconscious state at randomization have greatly higher odds of death compared to the drowsy reference. Patients who were fully alert at randomization had very low odds of death at six months compared to those who were drowsy or unconscious. Sex and aspirin treatment show relatively smaller effects, where aspirin was associated with a 6% reduction in odds, though this effect was not statistically significant ($p = 0.106$). Overall, these estimates quantify the relative impact of each covariate on the probability of death, allowing for a clear comparison of risk factors in the study population.

Parameter	2.5%	97.5%
Intercept	-5.002	-4.237
AGE	0.062	0.070
SEX (M)	0.010	0.163
RSBP	-0.005	-0.002
CONSCIOUS (F)	-1.576	-1.417
CONSCIOUS (U)	1.154	1.782
ASPIRIN (Y)	-0.136	0.013

Table 3: Normal-theory 95% confidence intervals for model coefficients.

Parameter	2.5%	97.5%
Intercept	-5.011	-4.229
AGE	0.062	0.070
SEX (M)	0.009	0.168
RSBP	-0.005	-0.002
CONSCIOUS (F)	-1.577	-1.421
CONSCIOUS (U)	1.156	1.780
ASPIRIN (Y)	-0.133	0.014

Table 4: Bootstrap 95% confidence intervals for model coefficients.

We sample from our dataset with replacement to form 1000 bootstrap samples. Table 4 shows the bootstrap 95% confidence intervals for our model parameters. For each parameter, the bootstrap 95% confidence interval was computed as the interval between the 2.5th and 97.5th percentiles of the coefficient estimates across the 1000 bootstrap resamples. Table 3 contains the 95% confidence intervals under normal theory assumptions which assumes the maximum likelihood estimates of the logistic regression coefficients are approximately normally distributed. The 95% confidence intervals were computed as $\hat{\beta} \pm 1.96 \text{SE}(\hat{\beta})$, where $\hat{\beta}$ is the estimated coefficient and $\text{SE}(\hat{\beta})$ its standard error. The bootstrap and normal-theory 95% confidence intervals for all model coefficients are very similar (Tables 4 and 3). This indicates that the normal-theory assumption for the logistic regression coefficients is reasonable in this dataset, and that the asymptotic normality approximation provides reliable interval estimates. Figure 3 shows the predicted probabilities of death at six months from the model for each combination of sex, conscious state at randomisation and aspirin treatment as covariates. Conscious state greatly affects predicted probabilities of death as we observe three distinct clusters of curves. Then looking more closely we can see the model gives males a slightly higher probability of death than females. Importantly, we observe that aspirin treatment has a small lowering in predicted probability of death.

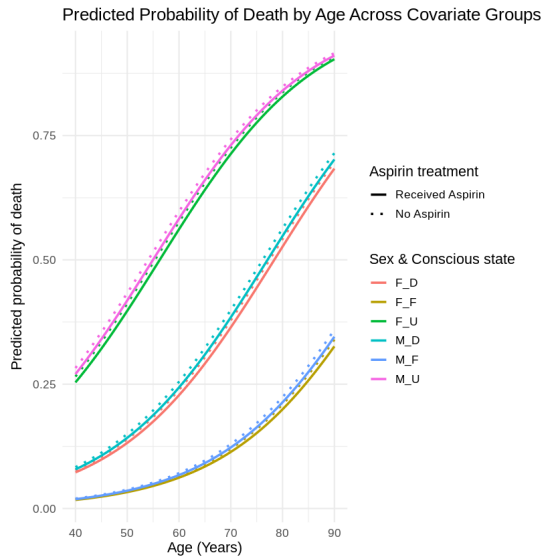


Figure 3: Predicted probability of death by age from the logistic regression model, shown for combinations of sex and conscious state.

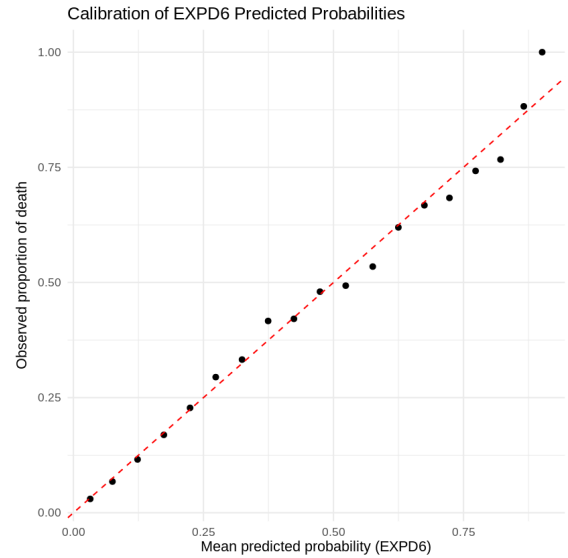


Figure 4: Predicted probability of death at 6 months vs observed proportion of deaths.

3 Power Analysis

IST_data contains under heading EXPD6 an expected probability of death at six months from randomisation. We will verify the given prediction is well calibrated. We do this by dividing the data by predicted probability into bins of width 0.05 between 0 and 1 then compute the mean predicted probability and observed proportion of deaths at six months for each bin. Figure 4 verifies the given predictions are accurate as the points fall along a line of slope 1. There is higher noise for the higher predicted probabilities due to smaller sample sizes, for example the last bin contains only one sample, but the lower predicted and higher sampled points fall very closely.

Figure 5 presents a box plot showing the distribution of predicted probability of death at six months for our Aspirin treatment groups. We find the distributions do not differ as in both median ≈ 0.160 , LQ ≈ 0.085 , UQ ≈ 0.320 .

With the predictive validity of EXPD6 and balanced treatment groups confirmed, we proceed to evaluate the power of the trial to detect a treatment effect by performing a simulation utilising and modifying the given predicted probabilities. The treatment effect was modeled as the odds ratio θ , where $1 - \theta$ is the proportion aspirin lowered odds of death.

We chose $\theta \in [0.8, 1.0]$, then transformed each patients baseline probabilities into odds, applied the treatment effect θ to the aspirin group, and transform back to modified probabilities. We generate binary outcomes based on these new probabilities and fitted a binomial Generalized Linear Model for 500 iterations per treatment effect to balance computational efficiency with estimate stability. Power was estimated as the proportion of simulations yielding a statistically significant treatment effect ($p < 0.05$). The results are visualised in Figure 6.

The simulation model was first validated under the null hypothesis ($\theta = 1.0$). The estimated power was 0.038 which aligns closely with our chosen significance level ($\alpha = 0.05$), confirming that the test has an appropriate Type I error rate.

Figure 6 shows a sigmoidal relationship between effect size and Power, this demonstrates high sensitivity for moderate treatment effects (a risk reduction of $\geq 14\%$), the power exceeds 97% making detection of such effects very high. A standard 80% power threshold has been overlaid. This is crossed between $\theta = 0.88$ (Power = 0.890) and $\theta = 0.90$ (Power = 0.756), indicating that the trial is sufficiently powered to detect a 12% reduction in the odds of death. However, statistical power declines rapidly for smaller effects; at $\theta = 0.94$, power falls to 0.310. This suggests that while the IST is well designed for moderate effects, it may lack the sensitivity to consistently confirm subtle benefits in the 5-10% range.

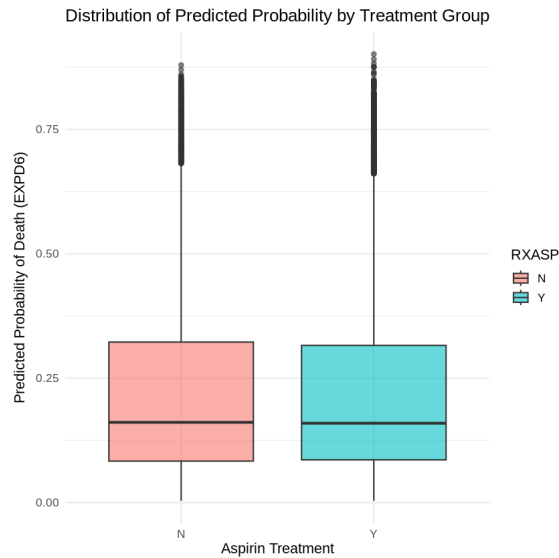


Figure 5: Distribution of predicted probability of death between treatment groups.

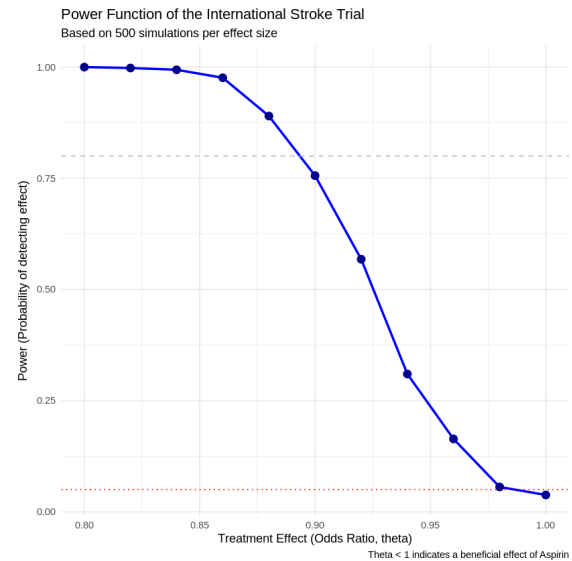


Figure 6: Power function of treatment effects.

4 Summary for Clinical Investigators

This analysis of the International Stroke Trial evaluated whether Aspirin reduces six month mortality. Although patients receiving Aspirin showed an estimated 6% reduction in the odds of death, this effect did not reach statistical significance at the 95% level, meaning the apparent benefit may be due to chance. Baseline clinical factors were far stronger predictors of mortality: unconscious patients had over four times the odds of death compared with drowsy patients, risk increased by roughly 7% per year of age, and higher systolic blood pressure was associated with slightly improved survival. The IST's design demonstrated successful randomisation ensuring balanced baseline characteristics, an extremely high follow up rate (< 0.8% unknown outcomes), and accurate calibration of the predicted probability of death (EXPd6), which closely matched observed death rates. However, simulated power analysis showed that the trial was only powered enough to detect more moderate treatment effects (> 12% risk reduction) and may miss smaller but still clinically meaningful benefits. Future research may therefore require larger sample sizes or combined evidence across studies to evaluate modest treatment effects for groups such as unconscious patients. As these groups had smaller sample sizes, stratifying randomisation would ensure enough data to prove whether the drug works for the sickest patients.

A R Code

Listing 1: Loading libraries, importing the dataset and performing cleanup

```
library(tidyverse)
library(dplyr)
library(ggplot2)
# Reading in the data and performing definitive cleanup
IST_data <- read_csv("IST-data.CSV") %>%
  filter(FDEAD %in% c("Y", "N")) %>% # Remove unknown outcomes
  mutate(
    FDEAD_bin = ifelse(FDEAD == "Y", 1, 0), # Create binary outcome (1 = Dead, 0 =
      Alive)
    SEX = factor(SEX), # Convert categorical predictors to
      factors
    CONSCIOUS = factor(RCONSC),
```

```

    ASPIRIN = factor(RXASP)
)

```

Listing 2: Code for Visualization of Joint Baseline Distribution

```

# Prepare data with proper labels and filter out NAs
grid_data <- IST_data %>%
  filter(!is.na(AGE) & !is.na(RSBP) & !is.na(RXASP) & !is.na(RCONSC) & !is.na(SEX))
  ) %>%
  mutate(
    aspirin_group = factor(RXASP,
                          levels = c("N", "Y"),
                          labels = c("Did Not Receive Aspirin", "Received Aspirin")
    ),
    conscious_level = factor(RCONSC,
                            levels = c("F", "D", "U"),
                            labels = c("Fully Alert", "Drowsy", "Unconscious")
    ),
    sex = factor(SEX,
                 levels = c("M", "F"),
                 labels = c("Male", "Female"))
  )

# --- Generate Scatter Plot with Linear Trend Lines ---
p_clean <- ggplot(grid_data, aes(x = AGE, y = RSBP, color = sex)) +
  geom_point(alpha = 0.15, size = 1) + # Use low alpha for large N
  geom_smooth(method = "lm", se = FALSE, linewidth = 1.5) + # Add linear trend (no
    CI band)

# Facet by Conscious Level (rows) and Aspirin Treatment (cols)
facet_grid(conscious_level ~ aspirin_group) +

scale_color_manual(values = c("Male" = "#4169E1", "Female" = "#FF69B4"),
                  name = "Sex") +

labs(title = "Age vs Systolic Blood Pressure by Treatment, Conscious Level, and
  Sex",
     x = "Age (years)",
     y = "Systolic Blood Pressure (mmHg)") +
theme_bw() +
theme(
  legend.position = "bottom",
  strip.text = element_text(size = 10, face = "bold"),
  strip.background = element_rect(fill = "grey90"),
  panel.grid.minor = element_blank()
)

print(p_clean)

```

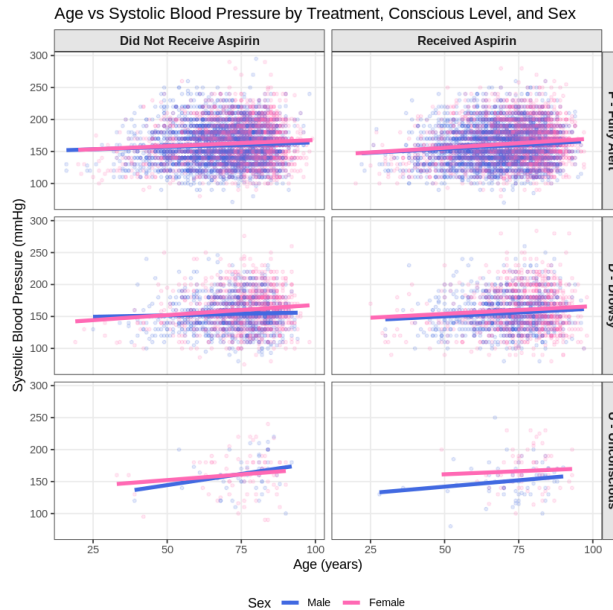


Figure 7: Joint distribution of age and baseline systolic blood pressure by treatment group and conscious state.

Listing 3: Calculation of Detailed Baseline Summary Table

```
# --- NOTE: IST_data is assumed to be loaded and cleaned from Listing 1 ---

# 1. Prepare data for grouping and define reporting factors
baseline_data <- IST_data %>%
  select(RXASP, AGE, SEX, RSBP, RCONSC) %>%
  mutate(
    # Create descriptive factor for reporting table
    aspirin_group = factor(RXASP, levels = c("N", "Y"),
                          labels = c("No Aspirin", "Aspirin")),
    # Ensure Sex and Conscious are treated as factors
    sex = factor(SEX),
    conscious_level = factor(RCONSC)
  )

# 2. Calculate summary statistics by treatment group
summary_stats <- baseline_data %>%
  filter(!is.na(aspirin_group)) %>%
  group_by(aspirin_group) %>%
  summarise(
    n = n(),
    mean_age = mean(AGE, na.rm = TRUE),
    sd_age = sd(AGE, na.rm = TRUE),
    mean_sbp = mean(RSBP, na.rm = TRUE),
    sd_sbp = sd(RSBP, na.rm = TRUE),
    n_male = sum(SEX == "M", na.rm = TRUE),
    n_female = sum(SEX == "F", na.rm = TRUE),
    pct_male = mean(SEX == "M", na.rm = TRUE) * 100,

    # Conscious State Counts (Using RCONSC original codes)
    n_alert = sum(RCONSC == "F", na.rm = TRUE),
    n_drowsy = sum(RCONSC == "D", na.rm = TRUE),
```



```

n_unconscious = sum(RCONSC == "U", na.rm = TRUE),

# Conscious State Percentages
pct_alert = mean(RCONSC == "F", na.rm = TRUE) * 100,
pct_drowsy = mean(RCONSC == "D", na.rm = TRUE) * 100,
pct_unconscious = mean(RCONSC == "U", na.rm = TRUE) * 100
)

print(summary_stats, width = Inf) # Output the full table

```

```

# A tibble: 2 x 15
  aspirin_group      n mean_age sd_age mean_sbp sd_sbp n_male n_female pct_male
  <fct>          <int>   <dbl> <dbl>   <dbl> <dbl> <int>  <int>   <dbl>
1 No Aspirin     9715    71.7  11.6    160.   27.5  5254   4461    54.1
2 Aspirin       9720    71.7  11.6    160.   27.7  5153   4567    53.0
# ... with 6 more variables: n_alert <int>, n_drowsy <int>, n_unconscious <int>,
#   pct_alert <dbl>, pct_drowsy <dbl>, pct_unconscious <dbl>
  n_alert n_drowsy n_unconscious pct_alert pct_drowsy pct_unconscious
  <int>   <int>       <int>       <dbl>   <dbl>       <dbl>
1   7458   2127         130       76.8    21.9         1.34
2   7463   2127         130       76.8    21.9         1.34

```

Listing 4: Calculating observed mortality proportions by subgroup

```

IST_summary <- IST_data %>%
  # Group by year-month, age, country
  group_by(RDATE, AGE, COUNTRY) %>%

  # Summarise counts
  summarise(
    total_n = n(),
    dead_n = sum(FDEAD == "Y"),
    prop_dead = dead_n/total_n,
    .groups = "drop"
  )

```

Listing 5: Computing mean and variance of proportions for model checking

```

# Compute the mean and variance of the proportion dead
# for each combination of AGE and COUNTRY
IST_mean_var <- IST_summary %>%
  group_by(AGE, COUNTRY) %>%
  summarise(
    mean_prop = mean(prop_dead), # average proportion dead within each group
    var_prop = var(prop_dead), # variance of the proportion dead within each
    group
    .groups = "drop"
  )

# Remove rows where variance could not be computed
IST_mean_var_filtered <- IST_mean_var %>%
  filter(!is.na(var_prop))

```

Listing 6: Visualization of Mean-Variance Relationship

```

# Initialize plot with data and map x/y axes
ggplot(IST_mean_var_filtered, aes(x = mean_prop, y = var_prop)) +
  geom_point(color = "blue", size = 2) + # Add scatter plot points (blue,
  size 2)

```

```

geom_smooth(method = "loess", span = 0.6,          # Add a smooth trend line (LOESS
  method)
            color = "red", se = FALSE) +          # Red line, disable confidence
            interval shading
labs(
  x = "Mean proportion of deaths",                # Label x-axis
  y = "Variance of proportion of deaths",          # Label y-axis
  title = "Scatter plot of Mean vs Variance"      # Add main chart title
) +
theme_minimal()                                  # Apply a clean, minimal
background theme

```

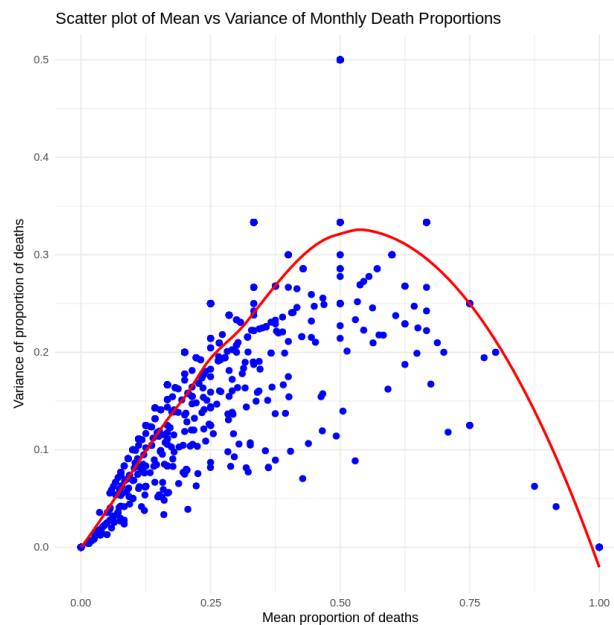


Figure 8: Mean proportion of deaths against variance of proportion of deaths.

Listing 7: Fit logistic regression model

```

glm_model <- glm( # Fit logistic regression model
  FDEAD_bin ~ AGE + SEX + RSBP + CONSCIOUS + ASPIRIN,
  data = IST_data,
  family = binomial(link = "logit")
)

summary(glm_model) # Display model summary and p-values
exp(coef(glm_model)) # Exponentiate coefficients to get Odds Ratios

```

Call:

```

glm(formula = FDEAD_bin ~ AGE + SEX + RSBP + CONSCIOUS + ASPIRIN,
    family = binomial(link = "logit"), data = IST_data)

```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-4.6198372	0.1952116	-23.666	< 2e-16 ***
AGE	0.0663563	0.0020705	32.049	< 2e-16 ***
SEX_M	0.0863457	0.0391034	2.208	0.0272 *

```

RSBP      -0.0032491  0.0006961  -4.668 3.05e-06 ***
CONSCIOUSF -1.4963721  0.0404011 -37.038 < 2e-16 ***
CONSCIOUSU  1.4678838  0.1604239   9.150 < 2e-16 ***
ASPIRINY   -0.0613626  0.0379946  -1.615  0.1063
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

(Dispersion parameter for binomial family taken to be 1)

```

Null deviance: 20635  on 19278  degrees of freedom
Residual deviance: 17344  on 19272  degrees of freedom
AIC: 17358

```

Number of Fisher Scoring iterations: 5

```

(Intercept)0.00985440018426557AGE1.06860742123918SEXM1.09018315013541RSBP0.996756196057396
CONSCIOUSF0.223941116262844CONSCIOUSU4.34004111642044ASPIRINY0.940482128730555

```

Listing 8: Initialization of Bootstrap Parameters and Storage

```

B <- 1000                                # Number of bootstrap replicates
n <- nrow(IST_data)                       # Sample size for resampling
p <- length(coef(glm_model))             # Number of model coefficients

bootstrap_estimates <- matrix(NA, nrow = B, ncol = p) # Initialize matrix to
  store coefficient estimates
colnames(bootstrap_estimates) <- names(coef(glm_model)) # Apply variable names to
  matrix columns

```

Listing 9: Execution of the Bootstrap Resampling Loop

```

set.seed(123) # for reproducibility

for (b in 1:B) {
  # Sample with replacement
  sample_idx <- sample(1:n, size = n, replace = TRUE)
  boot_data <- IST_data[sample_idx, ]

  # Fit logistic regression
  fit <- glm(FDEAD_bin ~ AGE + SEX + RSBP + CONSCIOUS + ASPIRIN,
    data = boot_data,
    family = binomial)

  # Store coefficients
  bootstrap_estimates[b, ] <- coef(fit)
}

```

Listing 10: Show Bootstrap Confidence Intervals

```

# Percentile method
CI_boot <- apply(bootstrap_estimates, 2, function(x) quantile(x, c(0.025, 0.975)))
CI_boot <- t(CI_boot) # transpose for readability
CI_boot

```

```

A matrix: 7 x 2 of type dbl
      2.5%      97.5%
(Intercept)-5.011416341-4.228682892
AGE          0.061975674 0.070584592

```

```

SEX      0.008571820 0.167958533
RSBP     -0.004630228 -0.001950449
CONSCIOUSF -1.576700509 -1.421082176
CONSCIOUSU 1.155668769 1.799834548
ASPIRINY -0.133093057 0.014384200

```

Listing 11: Show Normal-Theory Confidence Intervals

```

# Normal-theory method
normal_CI <- confint.default(glm_model)
normal_CI

```

```

A matrix: 7 x 2 of type dbl
      2.5%      97.5%
(Intercept) -5.002444863 -4.237229546
AGE          0.062298257 0.070414393
SEX          0.009704402 0.162987017
RSBP        -0.004613380 -0.001884773
CONSCIOUSF -1.575556838 -1.417187433
CONSCIOUSU 1.153458700 1.782308944
ASPIRINY    -0.135830589 0.013105325

```

Listing 12: Definition of Full Covariate Space for Prediction

```

# --- Define Full Covariate Space for Prediction ---

# Define age range
age_seq <- seq(40, 90, by = 1)

# Choose covariate levels
sex_levels <- c("M", "F")
conscious_levels <- c("F", "D", "U")
aspirin_levels <- c("Y", "N")

# Fix RSBP at mean
mean_RSBP <- mean(IST_data$RSBP, na.rm = TRUE)

# Create all combinations of covariates and ages
newdat <- expand.grid(
  AGE = age_seq,
  SEX = sex_levels,
  CONSCIOUS = conscious_levels,
  ASPIRIN = aspirin_levels,
  RSBP = mean_RSBP
)

```

Listing 13: Generating and Plotting Model Predictions Across Subgroups

```

# --- Format and Plot Predictions ---

# 1. Create plotting variables for visual mapping
newdat$linetype <- factor(newdat$ASPIRIN,
  levels = c("Y", "N"),
  labels = c("Received Aspirin", "No Aspirin"))

# Combine Sex and Conscious State for the color aesthetic
newdat$color_group <- factor(paste(newdat$SEX, newdat$CONSCIOUS, sep = "-"))

# 2. Calculate predicted probabilities from the model

```

```

# NOTE: Ensure 'glm_model' is used as the fit object
newdat$pred <- predict(glm_model, newdata = newdat, type = "response")

# 3. Plotting the predicted probability curves
# (ggplot2 is assumed loaded from Listing 1)
ggplot(newdat, aes(x = AGE, y = pred, color = color_group, linetype = linetype)) +
  geom_line(size = 1) +
  labs(
    x = "Age_Years",
    y = "Predicted_probability_of_death",
    color = "Sex_&_Conscious_state",
    linetype = "Aspirin_treatment",
    title = "Predicted_Probability_of_Death_by_Age_Across_Covariate_Groups"
  ) +
  scale_linetype_manual(values = c("solid", "dotted")) +
  theme_minimal(base_size = 13)

```

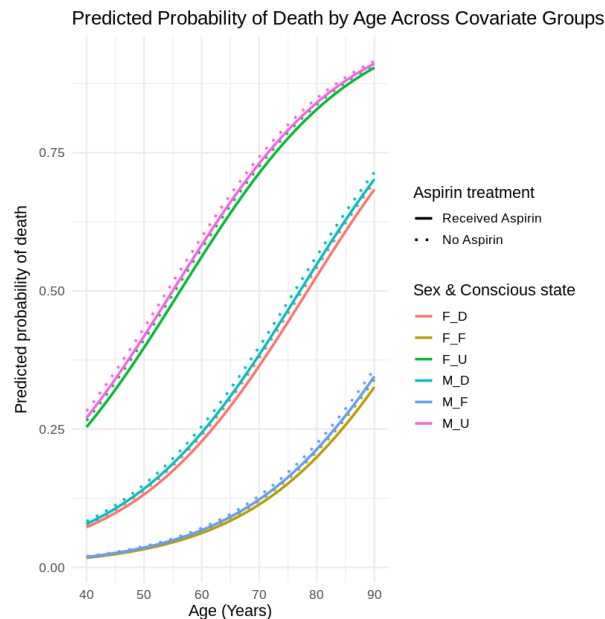


Figure 9: Predicted probability of death by age from the logistic regression model, shown for combinations of sex and conscious state.

Listing 14: Calculating and Plotting Model Calibration

```

# Set bin width for predicted probabilities
delta <- 0.05

# Bin predicted probabilities and calculate observed proportion
calibration <- IST_data %>%
  mutate(bin = cut(EXPD6, breaks = seq(0, 1, by = delta), include.lowest = TRUE,
    right = FALSE)) %>%
  group_by(bin) %>%
  summarise(
    mean_pred = mean(EXPD6, na.rm = TRUE), # mean predicted probability in
      the bin
    obs_rate = mean(FDEAD_bin, na.rm = TRUE), # observed proportion of death
    n = n() # number of individuals in the bin
  )

```

```

) %>%
ungroup()

# View the calibration table
print(calibration)

# Plot calibration
ggplot(calibration, aes(x = mean_pred, y = obs_rate)) +
  geom_point(size = 2) +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed", color = "red") +
  labs(
    x = "Mean_predicted_probability(EXPD6)",
    y = "Observed_proportion_of_death",
    title = "Calibration_of_EXP6_Predicted_Probabilities"
  ) +
  theme_minimal(base_size = 13)

```

A tibble: 19 x 4

	bin	mean_pred	obs_rate	n
	<fct>	<dbl>	<dbl>	<int>
1	[0,0.05)	0.0323	0.0301	2124
2	[0.05,0.1)	0.0750	0.0678	3863
3	[0.1,0.15)	0.123	0.116	3109
4	[0.15,0.2)	0.174	0.169	2258
5	[0.2,0.25)	0.224	0.228	1585
6	[0.25,0.3)	0.274	0.295	1134
7	[0.3,0.35)	0.324	0.333	908
8	[0.35,0.4)	0.374	0.416	694
9	[0.4,0.45)	0.424	0.421	608
10	[0.45,0.5)	0.474	0.480	552
11	[0.5,0.55)	0.524	0.493	507
12	[0.55,0.6)	0.575	0.535	507
13	[0.6,0.65)	0.625	0.620	418
14	[0.65,0.7)	0.675	0.668	361
15	[0.7,0.75)	0.723	0.684	275
16	[0.75,0.8)	0.773	0.742	225
17	[0.8,0.85)	0.821	0.767	133
18	[0.85,0.9)	0.866	0.882	17
19	[0.9,0.95)	0.901	1	1

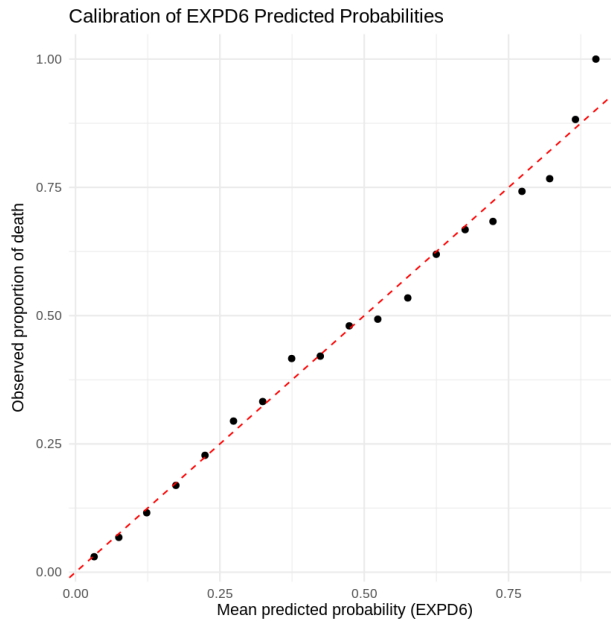


Figure 10: Predicted probability of death at 6 months vs observed proportion of deaths.

Listing 15: Assessment of Baseline Balance via Predicted Risk Distribution

```
# Boxplot to compare distributions
ggplot(IST_data, aes(x = RXASP, y = EXPD6, fill = RXASP)) +
  geom_boxplot(alpha = 0.6) +
  labs(
    x = "Aspirin_Treatment",
    y = "Predicted_Probability_of_Death(EXPD6)",
    title = "Distribution_of_Predicted_Probability_by_Treatment_Group"
  ) +
  theme_minimal(base_size = 13)

# Compare Distributions
IST_data %>%
  group_by(RXASP) %>%
  summarise(
    n = n(),
    median = median(EXPD6),
    Q1 = quantile(EXPD6, 0.25),
    Q3 = quantile(EXPD6, 0.75),
    IQR = IQR(EXPD6)
  )
```

```
# A tibble: 2 x 6
  RXASP      n median      Q1      Q3      IQR
<chr> <int> <dbl>   <dbl>   <dbl>   <dbl>
1 N      9644 0.1614 0.08350 0.32270 0.23920
2 Y      9635 0.1596 0.08600 0.31595 0.22995
```

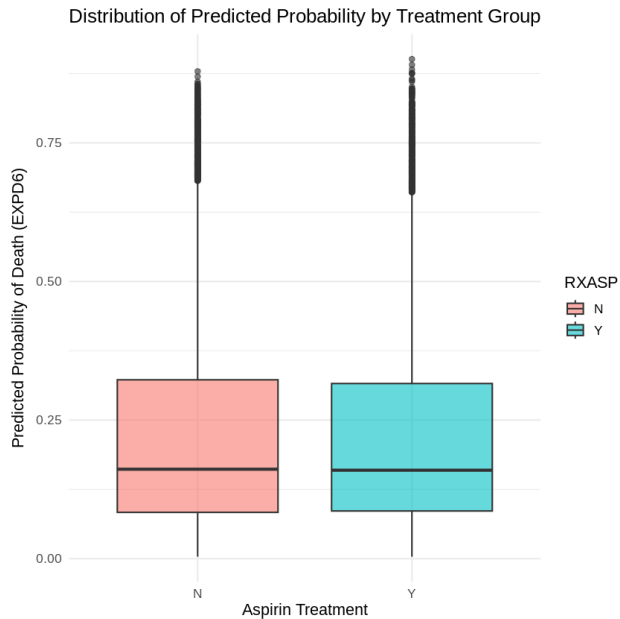


Figure 11: Distribution of predicted probability of death between treatment groups.

Listing 16: Execution of Power Analysis Simulation and Visualization

```
set.seed(123) # For reproducibility

# Filter for Aspirin vs No Aspirin and ensure complete data for EXPD6
sim_data <- IST_data %>%
  filter(RXASP %in% c("Y", "N")) %>%
  filter(!is.na(EXPD6)) %>%
  select(RXASP, EXPD6)

# theta represents the treatment effect
thetas <- seq(0.8, 1.0, by = 0.02)
n_sims <- 500 # Number of simulations per theta
alpha <- 0.05 # Significance level

# Initialize a data frame to store results
power_results <- data.frame(theta = numeric(), power = numeric())

# We loop through each Treatment Effect (theta)
for (theta in thetas) {

  significant_count <- 0

  # A. Calculate the 'True' Probability for this specific theta
  # We do this OUTSIDE the inner loop because it's the same for all sims at this
  # theta
  current_data <- sim_data %>%
    mutate(
      base_odds = EXPD6 / (1 - EXPD6),
      # Apply theta only if patient is in Aspirin group ("Y")
      new_odds = ifelse(RXASP == "Y", base_odds * theta, base_odds),
      # Convert back to probability
      true_prob = new_odds / (1 + new_odds)
    )
}
```



```

# B. Inner Loop: Run the trial n_sims times
for (i in 1:n_sims) {

  # rbinom(n, size, prob) generates 1s and 0s based on true_prob
  current_data$simulated_death <- rbinom(n = nrow(current_data), size = 1, prob
    = current_data$true_prob)

  # Fit Binomial GLM
  # We test if the model detects the difference between RXASP groups
  model <- glm(simulated_death ~ RXASP, family = binomial(link = "logit"), data
    = current_data)

  # Extract P-value for the treatment term
  # summary(model)$coefficients gives a matrix.
  # Row 2 is usually RXASPY (depending on factor levels), Column 4 is p-value.
  p_val <- summary(model)$coefficients[2, 4]

  # Check Significance
  if (p_val < alpha) {
    significant_count <- significant_count + 1
  }
}

# Calculate Power for this theta
calculated_power <- significant_count / n_sims

# Store result
power_results <- rbind(power_results, data.frame(theta = theta, power =
  calculated_power))

# Print progress
print(paste("Theta:", theta, "- Power:", calculated_power))
}

# Plotting the Power Curve
ggplot(power_results, aes(x = theta, y = power)) +
  geom_line(color = "blue", size = 1) +
  geom_point(size = 3, color = "darkblue") +
  # Add a reference line for 80% power (standard target)
  geom_hline(yintercept = 0.8, linetype = "dashed", color = "gray") +
  # Add a reference for 5% (Type I error rate at theta = 1)
  geom_hline(yintercept = 0.05, linetype = "dotted", color = "red") +
  labs(
    title = "Power Function of the International Stroke Trial",
    subtitle = paste("Based on", n_sims, "simulations per effect size"),
    x = "Treatment Effect (Odds Ratio, theta)",
    y = "Power (Probability of detecting effect)",
    caption = "Theta < 1 indicates a beneficial effect of Aspirin"
  ) +
  theme_minimal()

```

```

[1] "Theta: 0.8 - Power: 1"
[1] "Theta: 0.82 - Power: 1"
[1] "Theta: 0.84 - Power: 0.988"
[1] "Theta: 0.86 - Power: 0.96"
[1] "Theta: 0.88 - Power: 0.9"
[1] "Theta: 0.9 - Power: 0.754"

```

```

[1] "Theta: 0.92 - Power: 0.542"
[1] "Theta: 0.94 - Power: 0.314"
[1] "Theta: 0.96 - Power: 0.146"
[1] "Theta: 0.98 - Power: 0.078"
[1] "Theta: 1 - Power: 0.032"

```

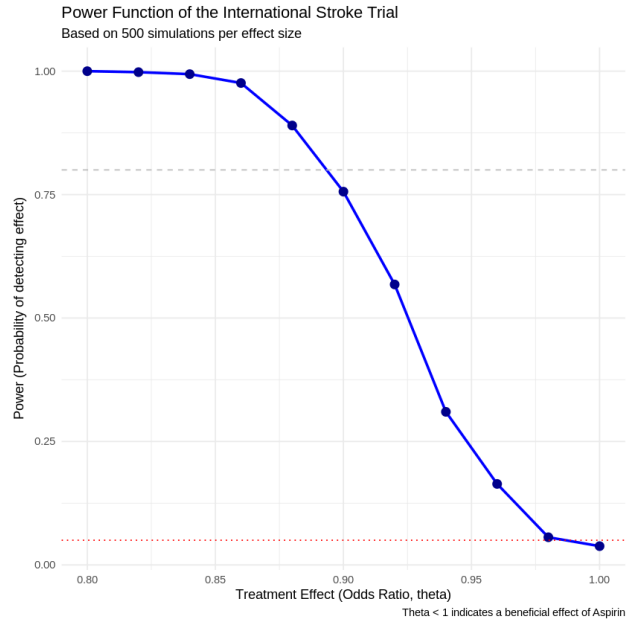


Figure 12: Power function of treatment effects.